

MUTUAL FUND ANALYTICS

Bluestock Fintech Internship

Final Project Report

End-to-end Data Analytics • ETL • SQL • EDA • Advanced Analytics • Tableau

Project scope	Industry, fund performance, investor and SIP/market analytics
Data	10 primary CSV datasets
Records	87,533
Database	SQLite
BI / Visualization	Tableau
Dashboard pages	4

Prepared for Bluestock Fintech project evaluation

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Document note

This report is intentionally source-grounded. Calculations and findings are based on the project datasets and completed analytical work. Values are not altered simply to match a reference value when the supplied source produces a different result.

1. Executive Summary

This capstone implements an end-to-end mutual fund analytics workflow covering **87,533 records across 10 primary datasets**. The project moves from raw CSV ingestion and validation through ETL preparation, SQLite modeling, exploratory data analysis, performance and risk analytics, and interactive Tableau reporting.

The final BI solution contains four dashboard pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. The dashboards were designed around the Bluestock task specification and include the requested KPIs, charts, slicers, tooltips and NAV-detail drill-through.

The analytical work combines industry measures such as AUM, SIP inflows and folios with fund-level return/risk measures, investor transaction behavior and category/portfolio concentration. This creates a single analytical narrative from industry-level trends to scheme-level performance and investor-level behavior.

Key source-derived findings include **■31,002 crore** of SIP inflows in December 2025, **26.12 crore** total folios in December 2025, strong growth in folios across the available series, and Banking as the largest sector exposure in the supplied holdings analysis at approximately **19.18%**.

The project also includes advanced analytics: historical 5% VaR/CVaR, rolling Sharpe, investor cohorts, SIP continuity, a risk-appetite recommender, sector HHI, benchmark comparison and tracking error.

A key reporting principle is transparency. The December 2025 source data yields **62.74 lakh crore** AUM and **1,522 schemes** in the dashboard calculations, which differ from the task-text reference values of 81L Cr and 1,908. The report preserves the source-derived values rather than fabricating them.

2. Project Objectives & Scope

Project objectives

- Integrate heterogeneous mutual fund datasets into a consistent analytical workflow.
- Validate schemas, identifiers, dates, numeric fields and transaction-level data before analysis.
- Build a reusable SQLite analytical layer.
- Perform EDA across industry, investor, SIP, category, portfolio and benchmark dimensions.
- Calculate return and risk metrics for scheme comparison.
- Extend the analysis with advanced risk and investor analytics.
- Deliver four interactive Tableau dashboards aligned to the Bluestock specification.
- Prepare professional documentation and final submission artifacts.

Scope mapping to the Bluestock task

Task area	Implemented work
Industry Overview	AUM KPIs, SIP inflows, folios, schemes, industry AUM trend, AUM by AMC
Fund Performance	Return vs risk scatter, sortable scorecard, NAV vs benchmark, slicers
Investor Analytics	State transaction amount, transaction-type donut, average SIP by age, monthly volume, slicers
SIP & Market Trends	SIP + NIFTY 50 dual axis, category heatmap, Top 5 FY25 categories
Interactivity	Tooltips, slicers and fund-table to NAV Detail drill-through

The scope intentionally excludes additional visuals that were not required by the task specification.

3. Data Sources & Data Dictionary

Dataset	Rows	Analytical purpose
01_fund_master.csv	40	Scheme master, category, plan, benchmark, expense and risk metadata
02_nav_history.csv	46,000	Daily NAV history for 40 schemes
03_aum_by_fund_house.csv	90	Fund-house AUM and scheme counts
04_monthly_sip_inflows.csv	48	Monthly SIP inflows, accounts and SIP AUM
05_category_inflows.csv	144	Category-level monthly net inflows
06_industry_folio_count.csv	21	Industry folio counts
07_scheme_performance.csv	40	Scheme returns, risk metrics, AUM and ratings
08_investor_transactions.csv	32,778	Investor transactions, demographics and geography
09_portfolio_holdings.csv	322	Holdings, sectors, weights and market values
10_benchmark_indices.csv	8,050	Daily benchmark index observations

Total: **87,533 records** across the 10 main datasets.

Key analytical fields

Domain	Important fields
Fund / scheme	Amfi Code, Scheme Name, Fund House, Category, Sub Category, Plan, Benchmark
NAV	Amfi Code, Date, Nav
AUM	Date, Fund House, Aum Lakh Crore, Aum Crore, Num Schemes
SIP	Month, Sip Inflow Crore, Active SIP Accounts Crore, New SIP Accounts Lakh
Investor	Investor Id, Transaction Date, Transaction Type, Amount Inr, State, Age Group, City Tier
Performance	1Y/3Y/5Y returns, benchmark return, alpha, beta, Sharpe, Sortino, volatility, drawdown
Holdings	Amfi Code, Stock Symbol, Sector, Weight Pct, Market Value Cr
Benchmark	Date, Index Name, Close Value

4. ETL Design & Data Quality

The ETL process was structured to preserve source meaning and analytical grain while making the datasets reliable for SQL and Tableau.

ETL pipeline

- **Extract:** Load all supplied CSV files and inspect structure, data types, dates and record counts.
- **Validate:** Check identifiers, dates, numeric fields, categorical values and expected row counts.
- **Transform:** Standardize fields, parse dates, create analytical date/month keys and prepare derived measures.
- **Model:** Store dimensions and fact tables in SQLite while retaining each dataset's natural grain.
- **Analyze:** Generate EDA, performance metrics and advanced analytics.
- **Visualize:** Connect the prepared data to Tableau and construct the four required dashboards.

Data-quality controls applied

- Row counts were checked across all 10 main datasets.
- Daily NAV was kept at scheme-day grain rather than being prematurely aggregated.
- Daily benchmark data was aggregated appropriately for monthly SIP comparison.
- Amfi Code was used for scheme-level relationships; Fund House was used for AUM relationships; Month was used for SIP/category/folio relationships.
- Source-derived dashboard values were retained where the supplied data differed from task-text reference values.

Important implementation detail

For the SIP + NIFTY 50 dual-axis view, the benchmark is daily while SIP data is monthly. The dashboard therefore uses **AVG(Close Value)** for the benchmark at monthly level; summing daily close values would create a misleading scale.

5. SQL Database Design

A SQLite database was created as a structured analytical layer for repeatable queries, validation and downstream analysis.

Object	Role
dim_fund	Fund/scheme master dimension
dim_date	Reusable date dimension
fact_nav	Daily NAV fact table
fact_aum	Fund-house AUM fact table
fact_transactions	Investor transaction fact table
fact_performance	Scheme performance and risk fact table

Typical analytical queries

- Rank funds by return and risk-adjusted metrics.
- Aggregate transaction value by state, age group and city tier.
- Summarize AUM by fund house and time period.
- Calculate transaction volumes by month.
- Validate dashboard KPIs independently from the visualization layer.

Database design principle

The database avoids forcing every dataset into one flat table. Preserving grain reduces duplication risk and makes analytical results easier to validate.

6. Exploratory Data Analysis — Industry & SIP

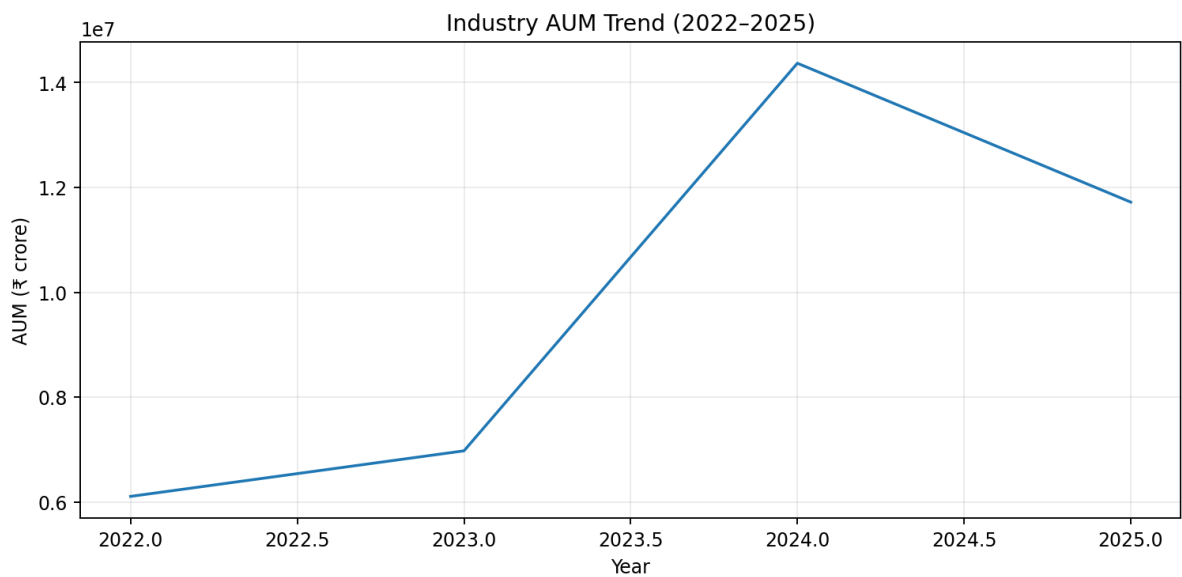


Figure 1. Industry AUM trend from the supplied AUM dataset.

The supplied AUM series shows a strong upward movement across the available 2022–2025 observations, with a pronounced 2024 value followed by a lower 2025 observation. This is a dataset-derived trend and is not presented as an external market estimate.

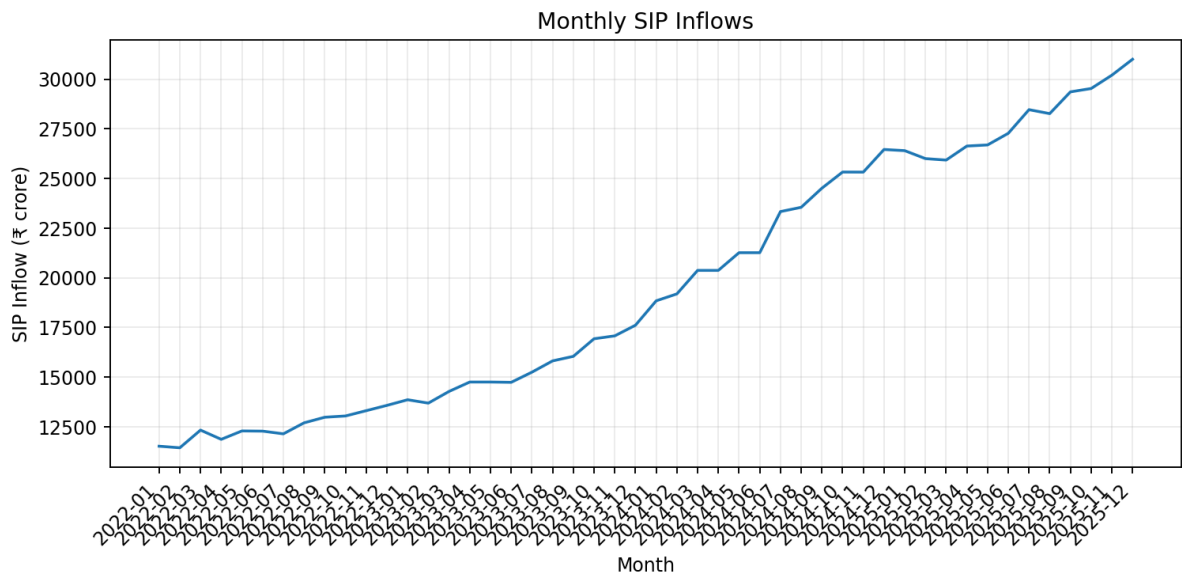


Figure 2. Monthly SIP inflows.

The highest supplied monthly SIP inflow is **31,002 crore in December 2025**. This value is used in the Industry Overview dashboard as the SIP inflow KPI.

6.1 Industry Scale — Folios & AMC AUM

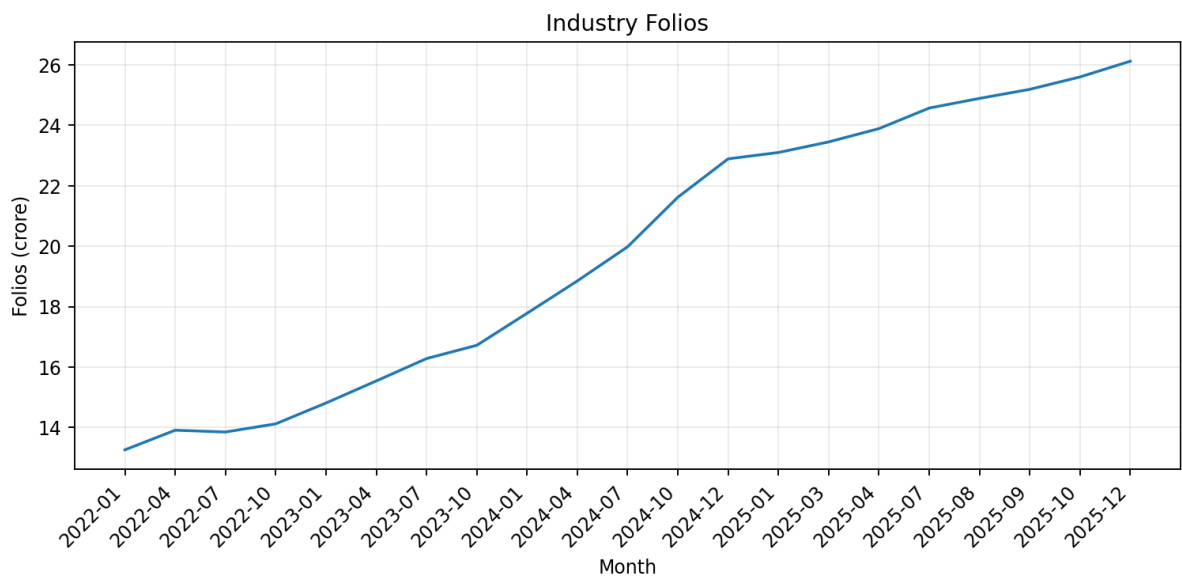


Figure 3. Industry folio trend.

Total folios increase from **13.26 crore in January 2022** to **26.12 crore in December 2025**, an increase of 12.86 crore or approximately 96.98% based on the supplied series.

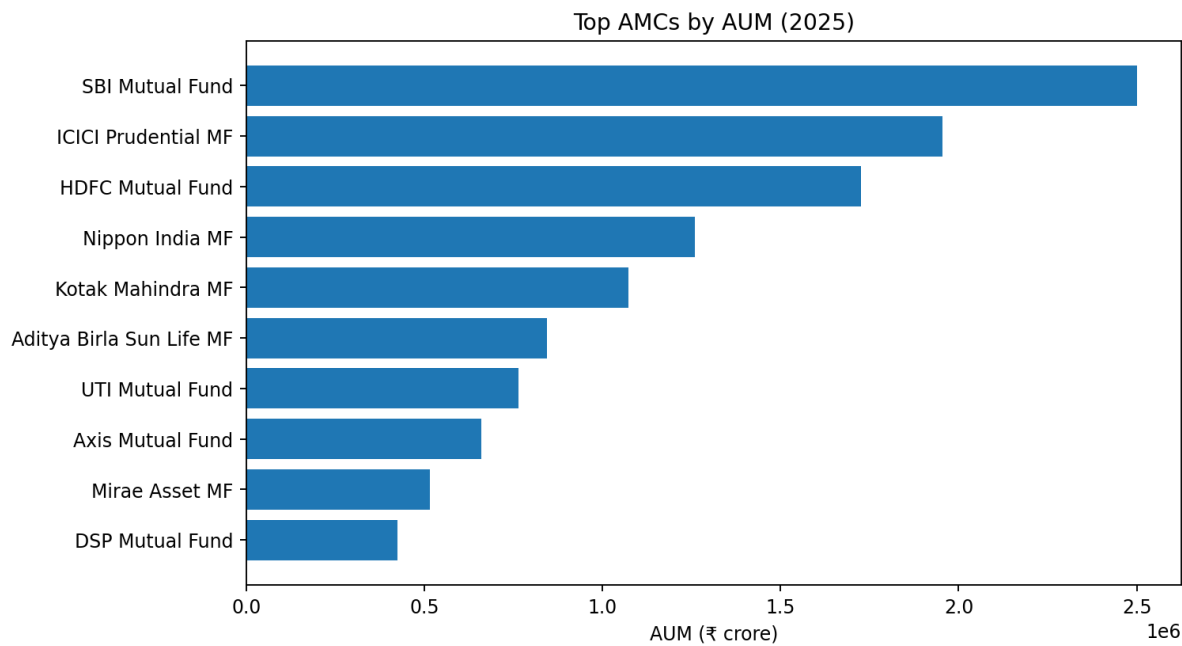


Figure 4. Top AMCs by AUM in the available 2025 AUM data.

The AMC view ranks fund houses by AUM and is used as the horizontal bar chart in the Industry Overview dashboard.

7. Investor Analytics

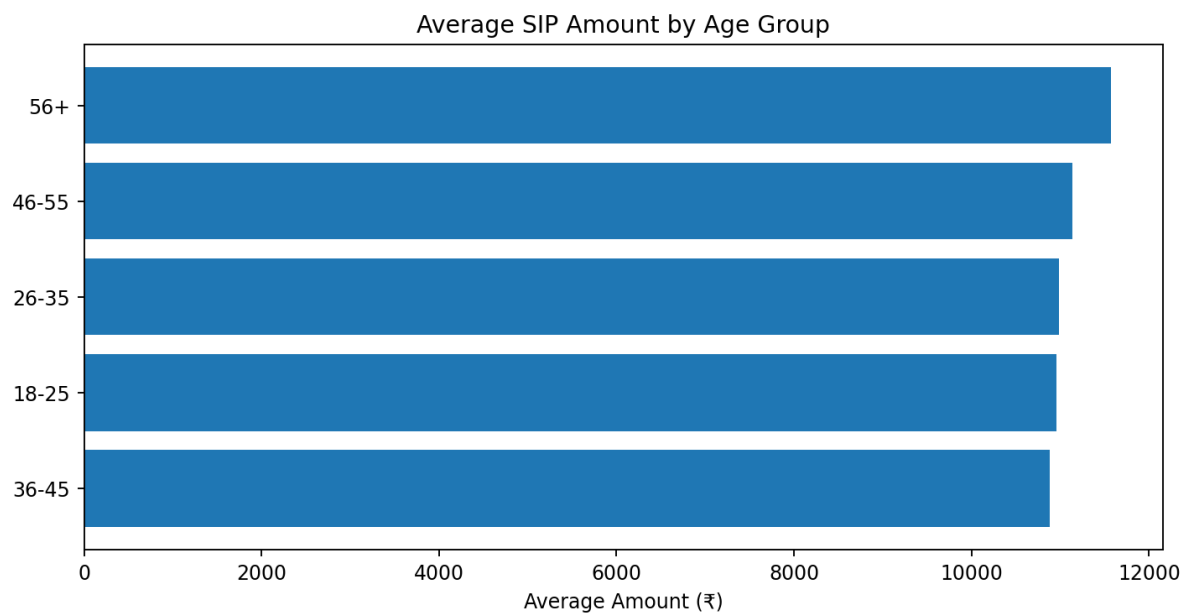


Figure 5. Average SIP transaction amount by age group.

This analysis filters transactions to **SIP** and compares average amount by age group. The highest supplied average is approximately **₹11,575** for the 56+ group.

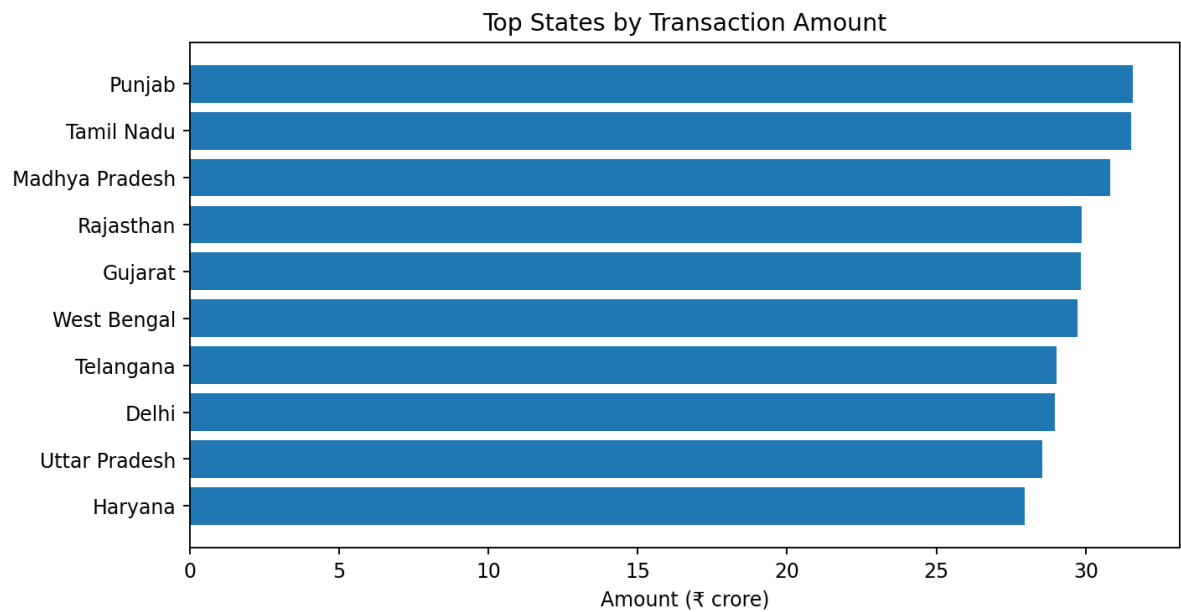


Figure 6. Top states by transaction amount.

State-wise transaction value is used in the Investor Analytics dashboard to identify geographic concentration.

7.1 Transaction Mix & Fund Performance

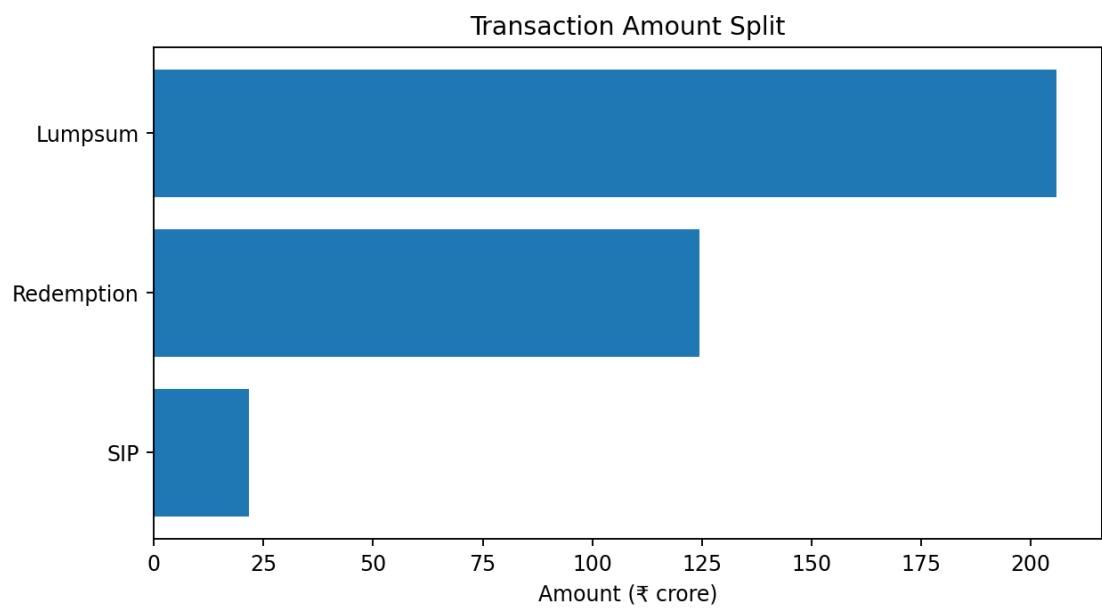


Figure 7. Transaction amount split across SIP, Lumpsum and Redemption.

The transaction-type analysis supports the required donut visualization. The dashboard displays the relative contribution of SIP, Lumpsum and Redemption amounts.

Fund performance overview

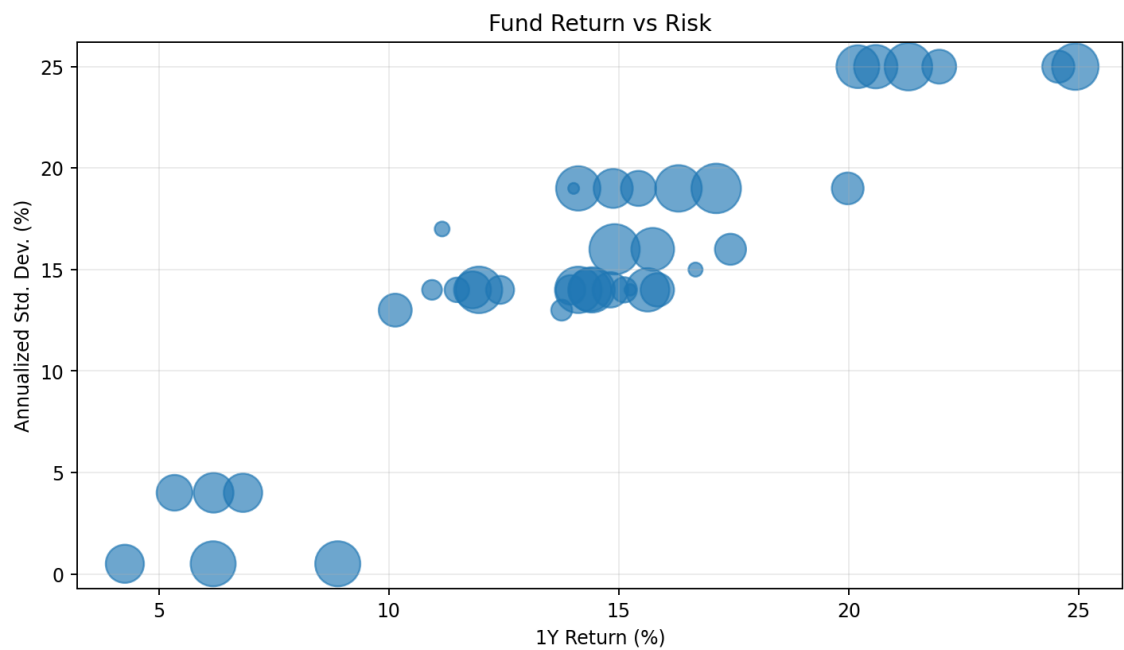


Figure 8. Fund return versus annualized risk; bubble size represents AUM.

The return-risk scatter compares 1-year return with annualized standard deviation and uses AUM as bubble size. This enables simultaneous assessment of return, risk and fund scale.

8. Fund Performance Analytics

The performance layer was designed to move beyond simple return ranking and incorporate risk-adjusted and benchmark-relative measures.

Metrics calculated

- 1-year, 3-year and 5-year return measures.
- Sharpe ratio using a 6.5% risk-free assumption and annualized daily-return scaling.
- Sortino ratio based on downside volatility.
- Alpha and beta versus the selected benchmark using regression-based analysis.
- Maximum drawdown.
- Tracking error for benchmark comparison.

Fund scorecard

The scorecard combines five ranking components: **30% 3Y return, 25% Sharpe, 20% alpha, 15% inverse expense ratio and 10% inverse drawdown**. It is a comparative screening framework rather than an investment recommendation.

NAV versus benchmark

The Tableau Fund Performance page includes a NAV line against the NIFTY 50 benchmark, with Fund House, Category and Plan slicers. The separate NAV Detail sheet provides continuous daily NAV history for drill-through from the scorecard.

9. Advanced Analytics & Risk Metrics

Analysis	Purpose
Historical VaR / CVaR (5%)	Estimate downside loss thresholds and expected tail loss
90-day rolling Sharpe	Monitor changes in risk-adjusted performance over time
Investor cohort analysis	Compare investor activity by transaction cohort
SIP continuity	Identify regular versus at-risk SIP continuity patterns
Risk-appetite recommender	Map selected fund characteristics to risk profiles
Sector concentration HHI	Measure concentration across portfolio sectors
Benchmark comparison	Compare fund performance with benchmark behavior
Tracking error	Measure deviation from benchmark returns

Selected completed findings

- The completed cohort analysis identified a 2024 cohort of 4,770 investors with approximately ₹225.81 crore in transaction value.
- SIP continuity analysis identified 1,362 eligible investors, with 1 regular and 1,361 classified as at-risk under the implemented continuity rule.
- Banking is the largest sector in the supplied holdings analysis at approximately 19.18%.
- Risk-appetite examples were generated for low and moderate profiles using the completed recommender logic.

10. SIP, Category & Portfolio Insights

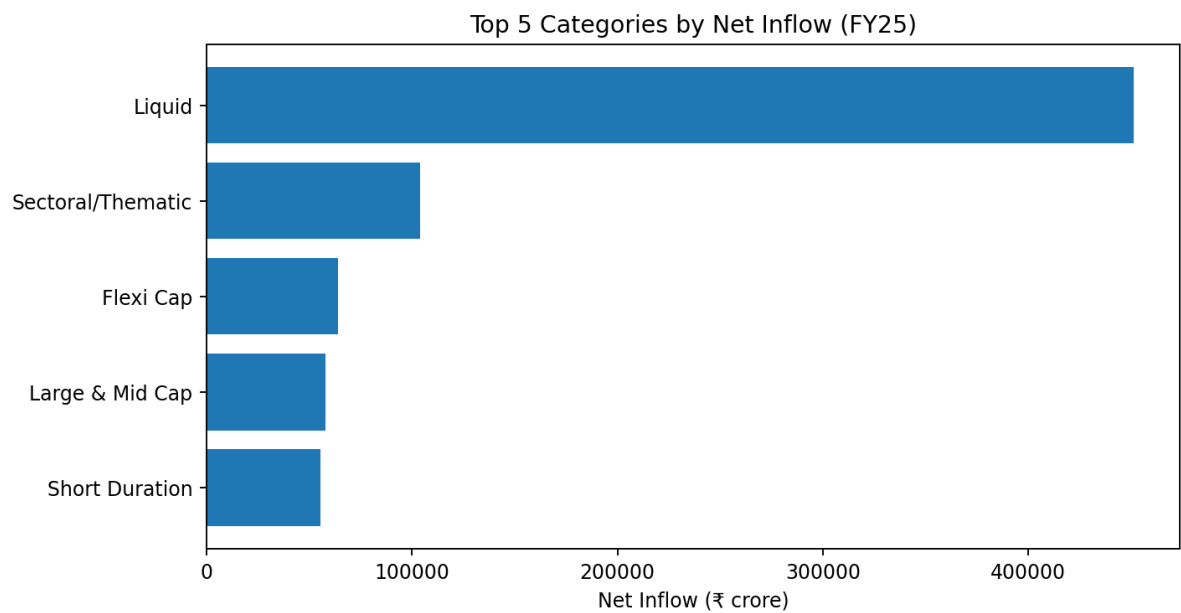


Figure 9. Top 5 categories by net inflow for the available FY25 category dataset.

The category dataset covers April 2024–March 2025. Within that available FY25 window, **Liquid** has the largest aggregate net inflow, followed by Sectoral/Thematic, Flexi Cap, Large & Mid Cap and Short Duration.

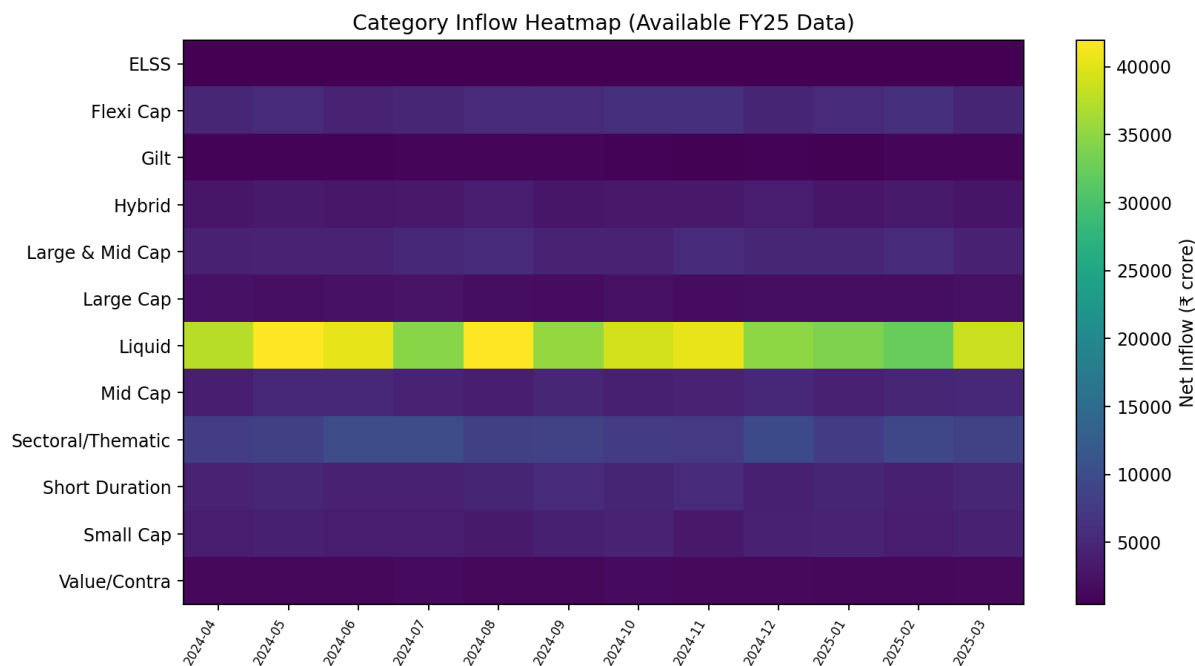


Figure 10. Category inflow heatmap using the available FY25 monthly category data.

The heatmap supports rapid identification of category-level inflow differences across months. Because the source contains 12 months, the analysis is not extended beyond the available FY25 period.

10.1 Portfolio Concentration & Tableau Implementation

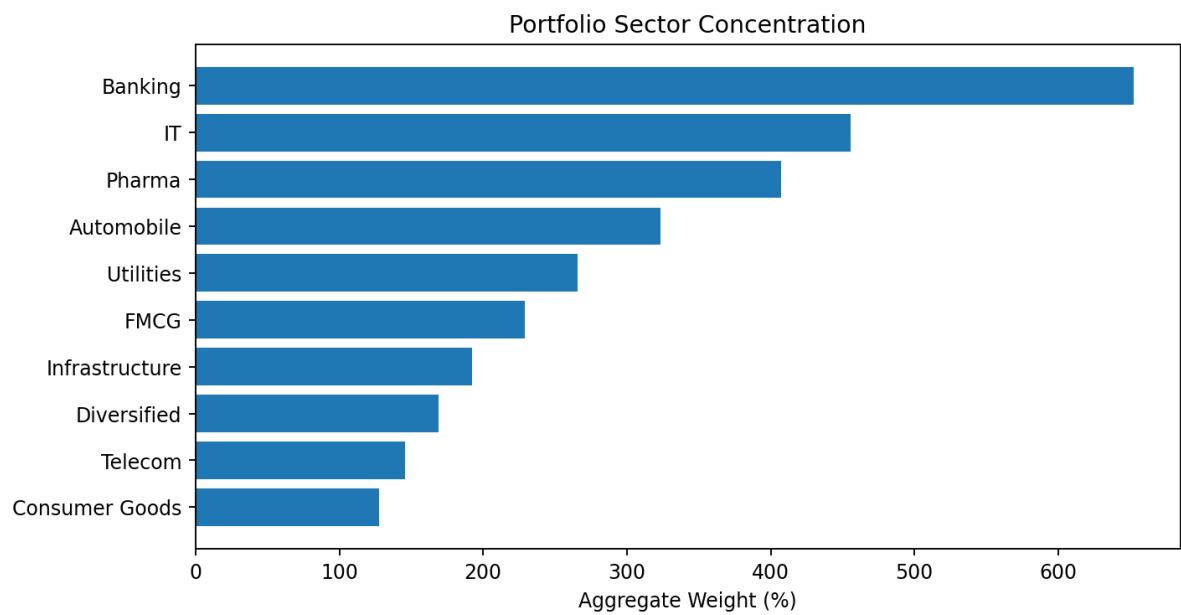


Figure 11. Portfolio sector concentration.

The portfolio holdings dataset contains 322 holdings across 14 sectors. Banking is the largest aggregate sector exposure at approximately 19.18%.

Tableau implementation

- **Industry Overview:** four KPI cards, AUM trend and AUM-by-AMC bar chart.
- **Fund Performance:** return-risk scatter, scorecard, NAV versus benchmark and three slicers.
- **Investor Analytics:** state amount bar, transaction-type donut, average SIP by age and monthly transaction volume, with three slicers.
- **SIP & Market Trends:** SIP/NIFTY 50 dual-axis chart, category heatmap and Top 5 FY25 category chart.
- **NAV Detail:** continuous daily NAV line used as the drill-through target from the fund scorecard.

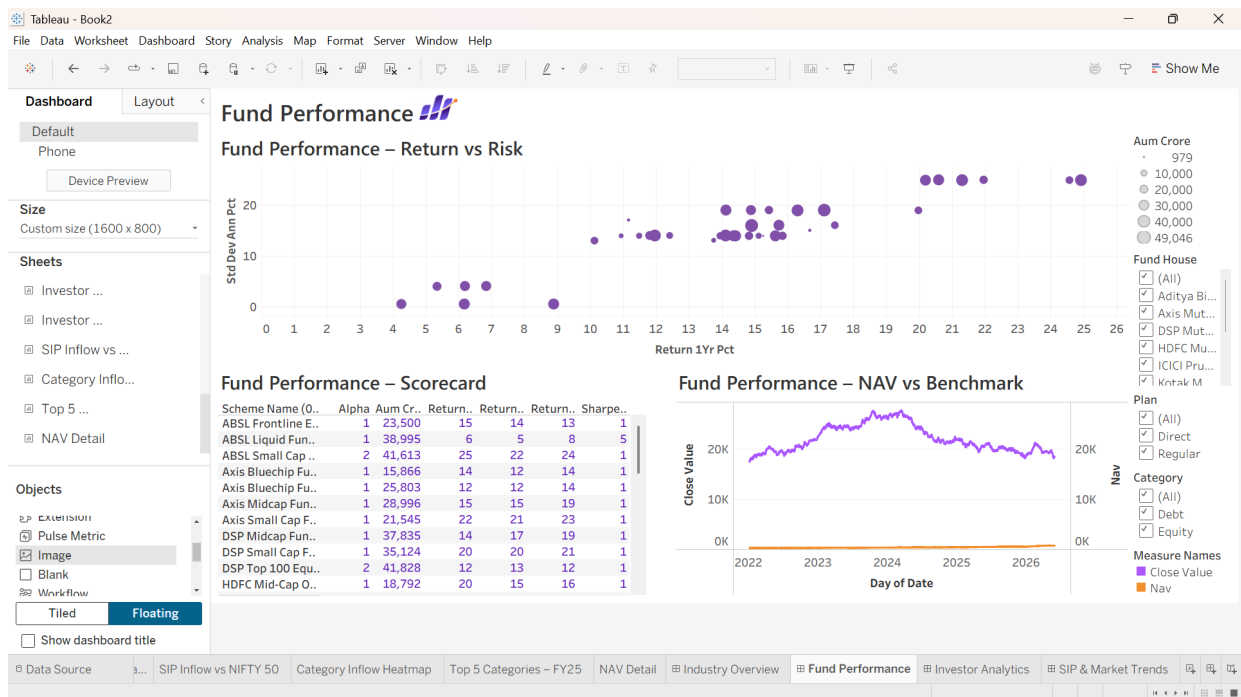
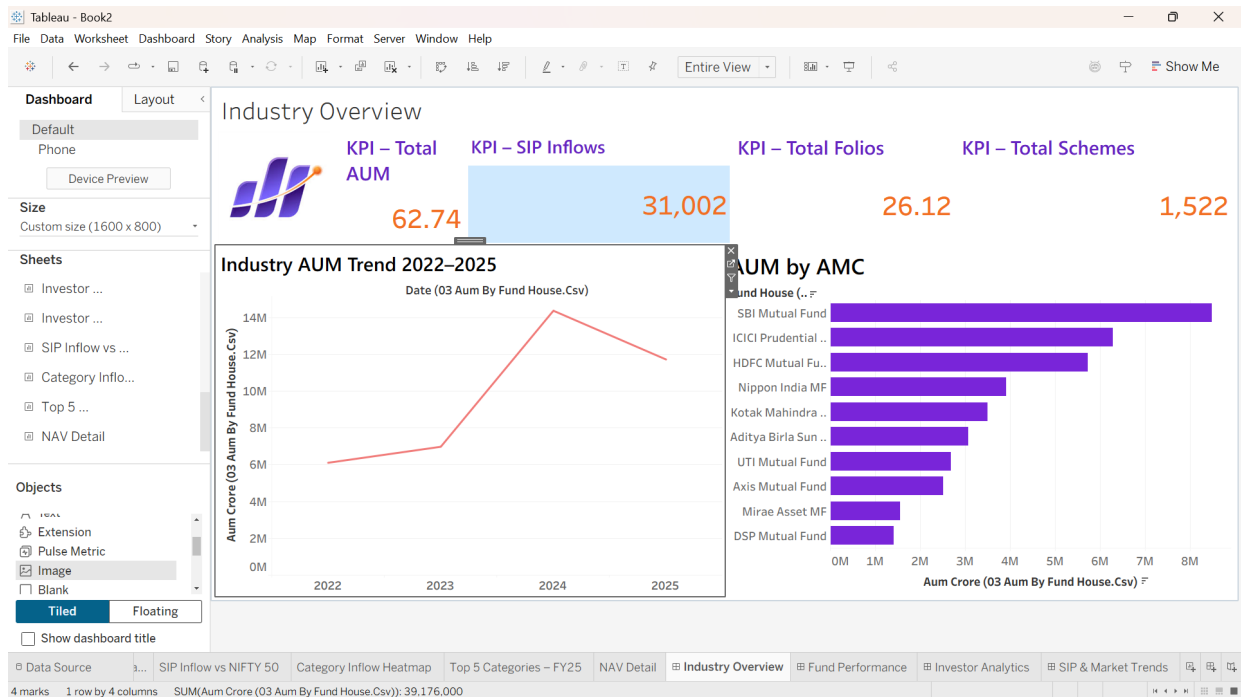
11. Dashboard Requirement Coverage

Bluestock requirement	Status	Implementation
Total AUM KPI	Completed	Source-derived December 2025 KPI: 62.74 lakh crore
SIP Inflows KPI	Completed	December 2025: ■31,002 crore
Folios KPI	Completed	December 2025: 26.12 crore
Schemes KPI	Completed	December 2025 source-derived count: 1,522
Industry AUM trend 2022–2025	Completed	Tableau line chart
AUM by AMC	Completed	Tableau horizontal bar chart
Return vs Risk scatter	Completed	Return X, Std Dev Y, AUM bubble size
Sortable fund scorecard	Completed	40 schemes with performance measures
NAV vs benchmark	Completed	NAV + NIFTY 50 dual-axis view
Fund House / Category / Plan slicers	Completed	Fund Performance dashboard
Transaction amount by State	Completed	Investor Analytics bar chart
SIP/Lumpsum/Redemption donut	Completed	Investor Analytics donut
Age Group vs Avg SIP Amount	Completed	SIP-filtered average amount
Monthly transaction volume	Completed	Monthly line chart
State / Age Group / City Tier slicers	Completed	Investor Analytics dashboard
SIP Inflow + NIFTY 50 dual axis	Completed	Monthly SIP bars + AVG NIFTY 50 line
Category inflow heatmap	Completed	Available FY25 category data
Top 5 categories by FY25 net inflow	Completed	Horizontal Top 5 chart
Drill-through to NAV Detail	Completed	Scorecard → NAV Detail filter/go-to-sheet actions
Tooltips	Completed	Configured on dashboard visuals
Bluestock logo/title treatment	Completed	Consistent dashboard branding

Data note

The task text lists reference KPI values of 81L Cr AUM and 1,908 schemes. The supplied project data produces 62.74 lakh crore AUM and 1,522 schemes for the December 2025 dashboard view. The report deliberately records the source-derived values.

12. Final Tableau Dashboard Screenshots



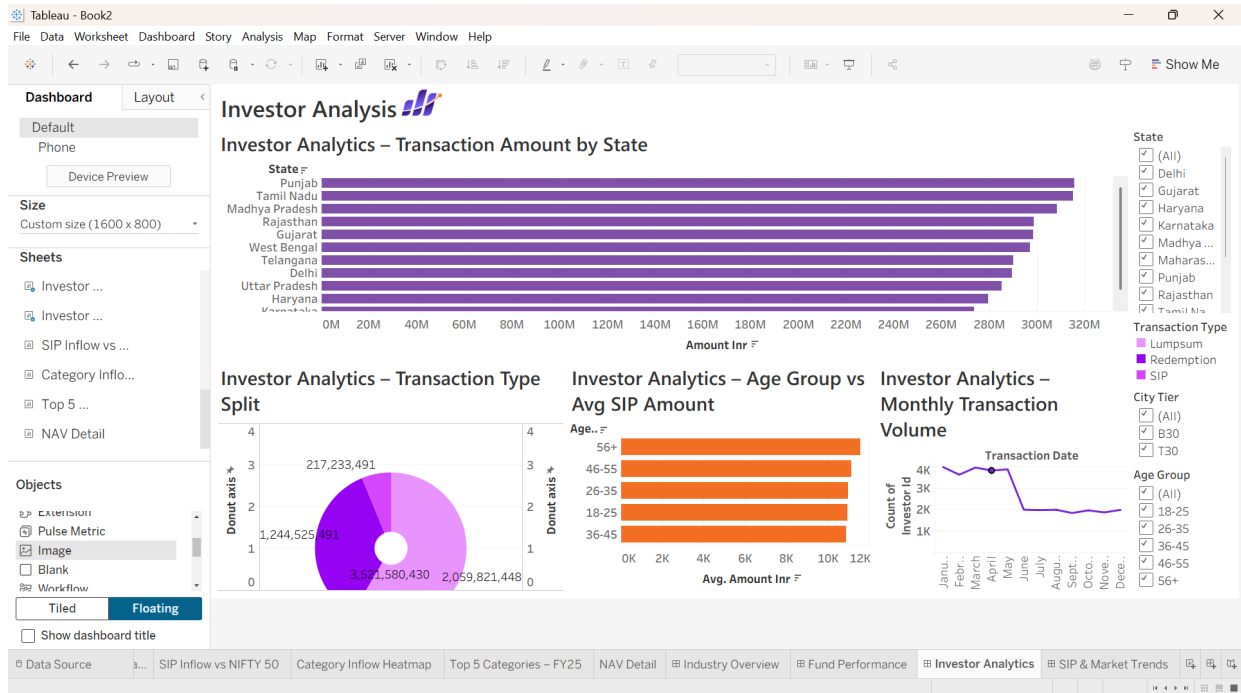


Figure 14. Investor Analytics — completed Tableau dashboard.

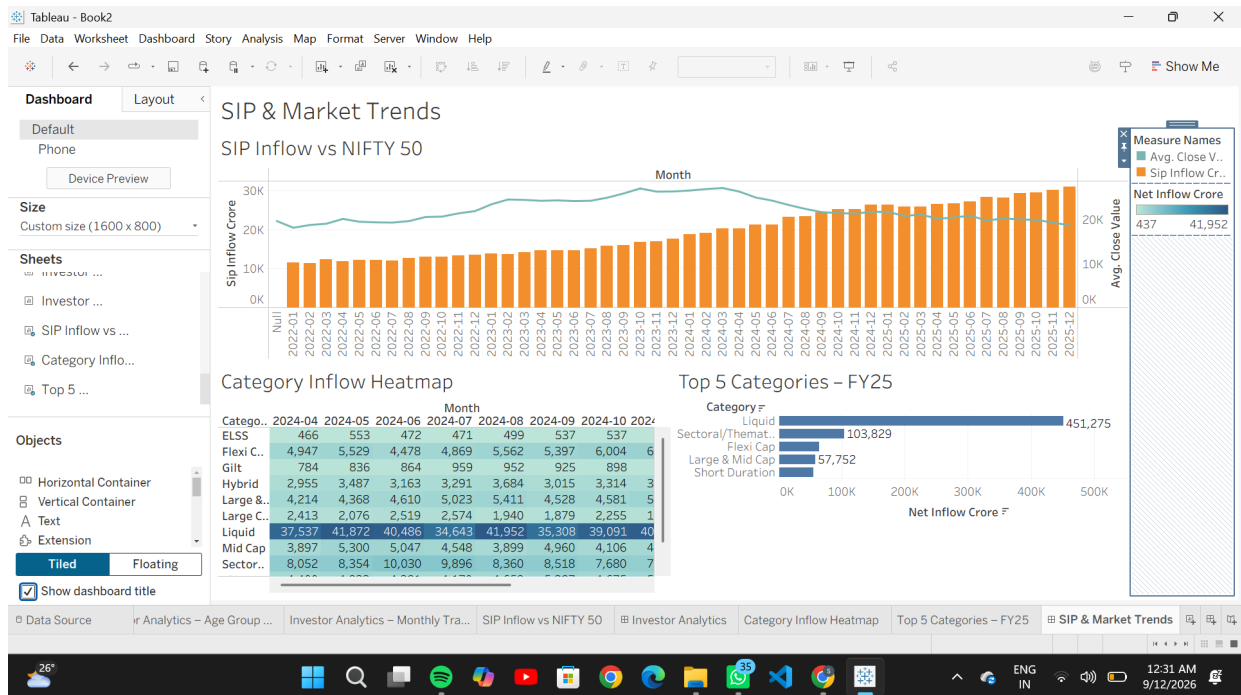


Figure 15. SIP & Market Trends — completed Tableau dashboard.

13. Limitations, Recommendations & Final Deliverables

Limitations

- Conclusions are limited to the supplied datasets and their available date ranges.
- The category inflow dataset covers 12 months, so FY25 is the appropriate window for the category heatmap and Top 5 category analysis.
- Historical analytics should not be interpreted as a guarantee of future performance.
- Dashboard KPI values are source-derived and are not altered to force agreement with task-text reference values.
- The task text uses '.pbix', which is a Power BI workbook format; the implemented dashboard is in Tableau.

Recommendations

- Use the fund scorecard together with risk and benchmark context rather than return alone.
- Monitor SIP inflow momentum and folio growth as complementary industry indicators.
- Track category and sector concentration for risk monitoring.
- Use age, geography and city-tier patterns to support investor segmentation.
- Refresh NAV, benchmark and transaction data periodically for continued monitoring.

Final deliverables checklist

Deliverable	Status
Final report PDF (15–20 pages target)	Completed — this 18-page professional report
Tableau workbook / dashboard implementation	Completed
4 dashboard pages	Completed
EDA and analytical outputs	Completed
SQLite database	Completed
README.md / data dictionary / SQL documentation	Completed
GitHub project history / final project code	Completed in project workflow
Presentation deck	Next task

Conclusion

The project demonstrates a complete analytics lifecycle: multi-source ingestion, data quality checks, structured SQL modeling, exploratory analysis, performance and risk analytics, advanced insights and interactive Tableau reporting. The final dashboard suite provides a practical view of mutual fund industry trends, scheme performance, investor behavior and SIP/market dynamics while remaining faithful to the supplied project data.

End of Report